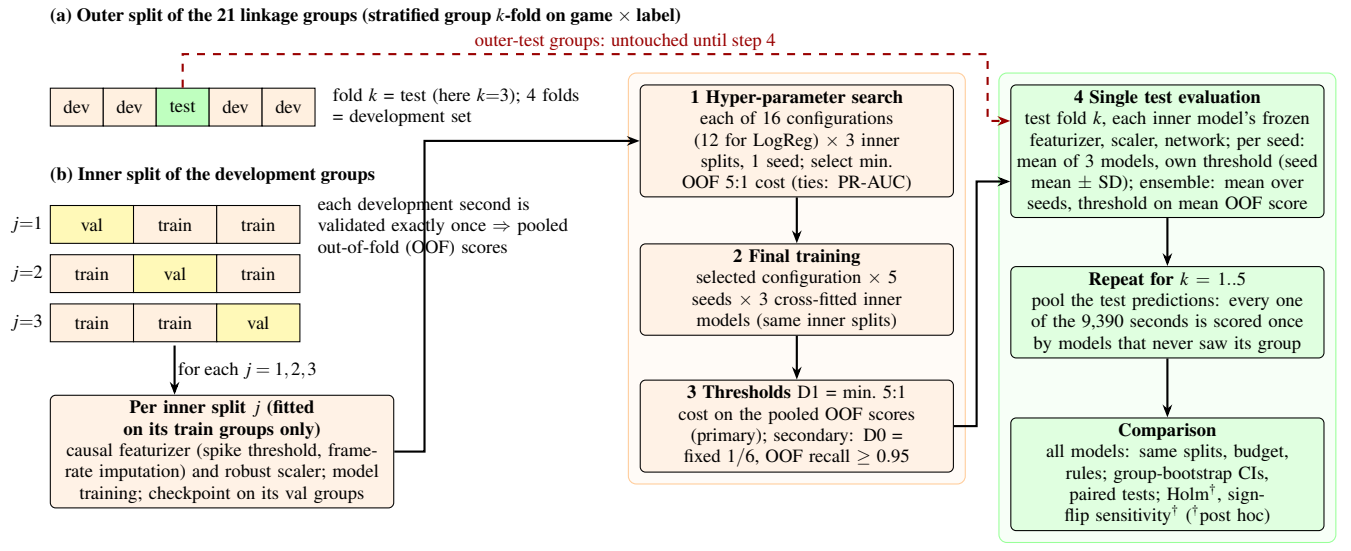

(a) Outer split of the 21 linkage groups (stratified group k-fold on game × label)
outer-test groups: untouched until step 4
dev
dev
test
dev
dev
fold k = test (here k=3); 4 folds = development set
(b) Inner split of the development groups
each development second is validated exactly once ⇒ pooled out-of-fold (OOF) scores
j=1
val
train
train
j=2
train
val
train
j=3
train
train
val
for each j = 1,2,3
Per inner split j (fitted on its train groups only)
causal featurizer (spike threshold, frame-rate imputation) and robust scaler; model training; checkpoint on its val groups
1 Hyper-parameter search
each of 16 configurations (12 for LogReg) × 3 inner splits, 1 seed; select min. OOF 5:1 cost (ties: PR-AUC)
2 Final training
selected configuration × 5 seeds × 3 cross-fitted inner models (same inner splits)
3 Thresholds D1 = min. 5:1 cost on the pooled OOF scores (primary); secondary: D0 = fixed 1/6, OOF recall ≥ 0.95
4 Single test evaluation
test fold k, each inner model's frozen featurizer, scaler, network; per seed: mean of 3 models, own threshold (seed mean ± SD); ensemble: mean over seeds, threshold on mean OOF score
Repeat for k = 1..5
pool the test predictions: every one of the 9,390 seconds is scored once by models that never saw its group
Comparison
all models: same splits, budget, rules; group-bootstrap CIs, paired tests; Holm†, sign-flip sensitivity† (†post hoc)